# The SPARK AI Consortium

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## POINT OF VIEW:
## Rethinking Enterprise Storage Architecture for AI-Driven Systems

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### INTRODUCTION

As AI initiatives move from pilot projects to production systems, they place new and sustained demands on enterprise storage infrastructure. Model training and inference require high-throughput access to large, diverse datasets and introduce I/O patterns that are iterative, parallel, and often unpredictable. These characteristics differ materially from the transactional and steady-state workloads for which many enterprise storage environments were originally designed.

Evidence from a broad survey of IT and storage professionals, supplemented by interviews with enterprise technology leaders, indicate that storage subsystems are frequently a limiting factor in AI deployments. Organizations report bottlenecks related to data movement, latency variability, fragmentation across tiers, and governance complexity. In this context, storage functions not merely as capacity, but as a performance-sensitive component of the AI stack. These findings suggest that storage architecture must be evaluated as a primary design variable in AI system performance rather than as an implementation detail.

Emerging AI-aware storage technologies—including predictive workload modeling, adaptive data tiering based on real-time access signals, metadata-centric indexing, and self-optimizing performance controls—portend an important shift towards more adaptive architectures. Drawing on over two decades of management experience in architecting storage systems design, engineering and enterprise strategy, we contend that production AI environments require storage layers capable of continuously evaluating data state and access behavior across the lifecycle. The primary challenge is architectural integration: aligning existing infrastructure with adaptive mechanisms that can sustain AI-scale throughput while maintaining policy compliance and cost-performance balance.

### AI IS TRANSFORMING THE CORE SUBSTRATES OF ENTERPRISE IT

Applying insights gained from our industry engagement, we conclude that AI is reshaping not only front-end applications but also the core substrate of enterprise IT—the systems that move, store, and govern data. AI-native storage technologies already provide capabilities like predictive workload modeling, dynamic tiering, metadata-driven indexing, and self-optimization. These capabilities enable organizations to better manage the explosive growth in volume, velocity, and variety characteristic of AI workloads. Storage infrastructure, once a stable and predictable layer, is becoming an active performance determinant in AI systems.

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The challenge to IT and storage system architects is how to implement AI-assisted evaluation of data wherever it is ingested, and to re-evaluate it periodically as the data's context, relevance, risk profile, and governance policies change. Drivers to this imperative include:

- Data quality shifts over time. Key data elements, lineage, and provenance can change by new conditions, regulations, or model requirements. Periodic reassessments are essential to detect degradation, bias, and drift.
- Storage tiering decisions cannot be static. Intelligent storage should continually reassess whether data belongs on high-performance, warm, or cold tiers as usage patterns shift.
- Regulatory and compliance requirements continue to evolve, especially under frameworks like GDPR, CPRA, and the EU AI Act. Storage policies must keep up by dynamically enforcing these constraints, continually rechecking metadata, consent, and risk classifications.
- AI systems themselves must be auditable, meaning stored data must be periodically evaluated for appropriateness, retention obligations, and explainability support.

Taken together, these drivers support the conclusion that storage modernization requires automated, AI-based "continuous assessment engines" to ensure that data remains properly governed, high-quality, and usable for future training, retraining, and validation.

## BEYOND THE BALANCE SHEET: AI'S IMPACT ON DATA VALUE

The ways in which companies have valued their data has long suffered an "accounting gap" whereby data has traditionally been underrepresented or ignored as a formal asset. AI can provide the refinery that can make that value measurable, systematic and actionable.

An important, nascent finding in our research is that AI can reverse the traditional assumption that all data loses value as it ages. Respondents in our interviews, and several industry surveys cited use cases where AI training data can increase in value the longer it is kept, because:

- Models require retraining, tuning, and evaluation against historical datasets.
- Archived data becomes an asset for auditability, model traceability, and regulatory review, including future requirements that may demand reconstruction of model decisions years later.
- Organizations reuse data across multiple models, not just the one the data was originally collected for.
- Synthetic datasets, prompt-engineering artifacts, model logs, and intermediate outputs—none historically considered "records"—now require long-term storage for legal and operational reasons.

On data retention, IDC reports that more than 40% of the organizations surveyed were planning for longer data retention periods and are redesigning tiered storage and archiving systems accordingly.[1] What is the effect of longer retention on AI model quality? In 2025, Recon Analytics reported that of the companies surveyed who had adopted AI, 90% believed that longer data retention improved the quality of AI outcomes, largely due to their ability to refine models using checkpoints.[2] Taken together, these findings suggest a measurable shift in the conventional "data value decays over time" model. In AI contexts, historical and semi-structured data can retain latent value as training, fine tuning, or model validation, measurably extending its economic half-

[1] A. Wright, G. Ireland and E. Burns, IDC (2025). *Content Creation in the Age of Generative AI: Implications for Value and Scale.*
[2] D. Kellogg, Recon Analytics (2025). *Impact of AI on Storage Requirements.*

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life. Resultingly, both industry surveys concluded that storage tiers traditionally positioned as compliance-driven or deep archives, such as tape, are being reevaluated as potential long term training repositories, designed for possible reactivation and computational reuse. AI is driving a paradigm shift in data retention policies and economics, blurring the line between temporary and long term data archives and permanent training repositories.

## DATA PRESERVATION AS RISK MANAGEMENT: REGULATORY AND LITIGATION DRIVERS OF STORAGE GROWTH

From an AI governance perspective, expanding regulatory regimes are materially increasing corporate data preservation obligations and litigation exposure. Frameworks such as the General Data Protection Regulation, the Sarbanes–Oxley Act, and enforcement oversight by agencies including the U.S. Securities and Exchange Commission require demonstrable controls over data retention, chain of custody, audit trails, and timely production. Concurrently, heightened eDiscovery expectations and regulatory investigations expand the scope of potentially discoverable electronically stored information (ESI). As a result, organizations face increasing legal risk if storage systems cannot support defensible retention, preservation holds, and policy-driven disposition—making storage architecture a critical component of enterprise risk management. Key storage policy considerations include:

- Proof of data consent and provenance
  - Verifiable records establishing lawful basis for processing, source validation, and chain of custody across internal and third-party datasets.
- Explainable model lineage
  - Documented traceability of dataset versions, preprocessing steps, model iterations, and deployment history to support accountability and reproducibility.
- Rich audit trails
  - Tamper-evident logs of data access, training activity, system changes, and inference events to enable regulatory review and internal investigations.
- Long-term retention of training data, intermediate outputs, and model weights
  - Preservation of training corpora, checkpoints, model versions, and tuning artifacts to support audits, disputes, and regulatory inquiries.
- Regulatory reporting and incident disclosure artifacts
  - Retention of investigation records, supervisory communications, incident reports, and compliance filings to demonstrate timely and accurate disclosure.
- Litigation hold and eDiscovery preservation for AI systems
  - Capability to preserve versioned datasets, model states, and related metadata in response to anticipated or active litigation.
- Data deletion workflows addressing persistent model influence
  - Defensible processes for operationalizing erasure rights while accounting for personal data potentially embedded in trained model parameters.

In short, storage architectures must embed governance at the storage layer itself. Modern storage isn’t just a performance engine—it’s also a governance engine.

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## LEAD PRACTICES: AI-RISK–AWARE STORAGE ARCHITECTURE

In our survey and interviews, we also asked respondents which leading practices guide their storage strategies. Six practices were consistently cited:

- AI-specific data classification, tied to model type and use case.
- Tiered retention frameworks balancing regulatory deletion with long-term reuse needs.
- Containerized, metadata-rich, AI-optimized storage systems with built-in data orchestration.
- Cross-functional AI risk and governance boards involving legal, compliance, data engineering, and ML teams.
- Continuous observability across hybrid and edge environments, enabling predictive scaling and performance tuning.

Our conclusions from this list are that leading organizations recognize that AI success is inseparable from storage modernization, and that storage modernization is inseparable from AI-guided, continuous evaluation of data quality, risk, and value.

## CONCLUSION: AI REQUIRES A PARADIGM SHIFT IN STORAGE STRATEGY

Based on our industry experience, our conclusion is clear: AI is transforming storage from a background infrastructure utility into a continuously adaptive, context aware system - a business-critical asset - requiring senior leadership. The imperative for organizations is to modernize their storage infrastructures not just to handle data growth, but to ensure that data is continuously assessed, optimally placed, properly governed, and preserved to serve evolving AI needs.

In short: AI data will be retained longer. Its value will increase. And only AI-aware storage architectures and clear strategic policies can support the technical, legal, and operational demands of the AI era.

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**About the SPARK AI Consortium**
SPARK AI was founded in 2024 as a response to the rapidly evolving AI landscape and the growing need for interdisciplinary collaboration to address the complex challenges of AI implementation in business contexts.

Hosted at the San Diego Supercomputer Center (SDSC), SPARK AI brings together the technical expertise of Center AI researchers in advanced computation, data science, and integrative AI solutions with the expertise, experience, and practical insights of industry leaders to create a unique ecosystem for innovation and thought leadership.

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